

same maximum-likelihood recipe

Chapter 1

$y \in \mathbb{R}$
continuous
target

$y \mid x \sim \mathcal{N}(\hat{y}, \sigma^2)$
identity link

$-\sum_i \log p(y_i \mid x_i)$

MSE
 $\sum_i (y_i - \hat{y}_i)^2$

Chapter 2

$y \in \{0, 1\}$
binary target

$y \mid x \sim$
Bernoulli($\hat{p}$)
sigmoid link

$-\sum_i \log p(y_i \mid x_i)$

binary cross-entropy
 $-\sum_i [y_i \log \hat{p}_i + (1 - y_i) \log (1 - \hat{p}_i)]$